



Research Design

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What is Research Design ?

- o A research design is a basic plan that guides the type of information to be collected, the source of data and phase of analysis of the research project.
- o The research design constitutes the blueprint for the collection, measurement, and analysis of data.
- o We can also say overall planning before doing investigation.

Classification of Research Design

- I. Survey method
- II. Exploring study
- III. Descriptive research design
- IV. Experimental research design
- V. Time series

Survey Method:

A survey is a method of gathering information from a number of individuals. This method is very much popular among researchers for collecting and analyzing facts, opinions, and interpretations of respondents, individuals or organizations that possess the relevant information.

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Types of Survey:

- Online survey.
- Paper survey.
- Telephonic surveys.
- One-to-one interviews.

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Exploratory Study:

- o The exploratory research is concerned with the discovery of a general form of the problem. Such kind of research requires flexibility. Usually researchers works informally but use tentative guideline.
- o This survey is used to know about new ideas and relationships.

Example:

- Tribal people in Bangladesh have some peculiar and uncommon background characteristics that distinguish them from the rest of the population.
- Random information suggests that they have a large family size, low age at marriage, and high mortality.
- The planned study is explorative, which will help the researchers to formulate objectives and hypotheses, keeping in view the above peculiarities regarding their distinct demographic characteristics.

Descriptive Research Design:

- A descriptive study is undertaken in order to ascertain and to be able to describe the characteristics of variables in a situation.
- The goal of a descriptive study, is to describe relevant aspects of the phenomena of interest to the researcher from an individual, organization, industry or other perspective.

Example:

- For example, in a supermarket, a researcher can from afar monitor and track the customers' selection and purchasing trends.

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Experimental Research Design:

Experimental designs are set up to study possible cause → effect relationships among variables. These are in contrast to correlational studies, which examine the relationships among variables without necessarily trying to establish which causes another.

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Example:

- An example of an experimental research design might be to investigate the effects of a new medication on blood pressure. In this study, participants would be randomly assigned to two groups: a treatment group that receives the new medication, and a control group that receives a placebo. The independent variable would be the medication (or placebo), while the dependent variable would be the participants' blood pressure.

Time Series:

A time series is a set of data points collected over a period of time, typically at regular intervals. Time series data is often used to analyze trends, patterns, and relationships between variables over time.

For example:

- Stock prices.
- Weather data.
- Economic indicators.
- Sales data

Research Design

Three steps are to be noted in research design.

1. What is a research design
2. Basic issue of a research design
3. Components of a research

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1. What is a research Design:

Research design is always necessary before the research is conducted.

- Good research design
- Objectivity
- Validity
- Reliability
- Generalization

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2. Basic issue of a research Design:

There are three basic issues of a research design:

- Type of investigation
- Purpose of the study
- Study setting

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Type of investigation:

Casual: When the researcher wants to describe the cause of the problem, then the study is called a casual study.

Non-causal (correlational study): When the researcher is interested in describing the important variables that are associated with the problem then it is known as non-causal or correlational study.

Purpose of the study:

Exploratory: The exploratory research is conducted with the discovery of a general form of the problem. Exploratory studies are necessary since we do not currently know if there are differences in common styles, interpretation scheme, superior-subordinate relation expectations, and the skills.

Descriptive: Descriptive study analysis the characteristics of availability in particular situation.

Purpose of the study:

Hypothesis Testing: Hypothesis testing means study in hypothesis, which explain certain relationship among the different variable. In other words, it is a procedure to test the statement about the population on the basis of sample information.

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Study setting:

There are two types of study setting.

Non-contrived study setting: When the study is conducted in natural environments, where events normally occur, is non-contrived settings. Correlational studies are conducted in non-contrived settings.

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Study setting:

Contrive study setting: In research, a contrived study setting is a research environment that has been intentionally created or manipulated by the researcher to observe a specific phenomenon or behavior under controlled conditions.

Contrived study settings are often used in laboratory-based research where the researcher has control over the variables that may affect the outcome of the study.

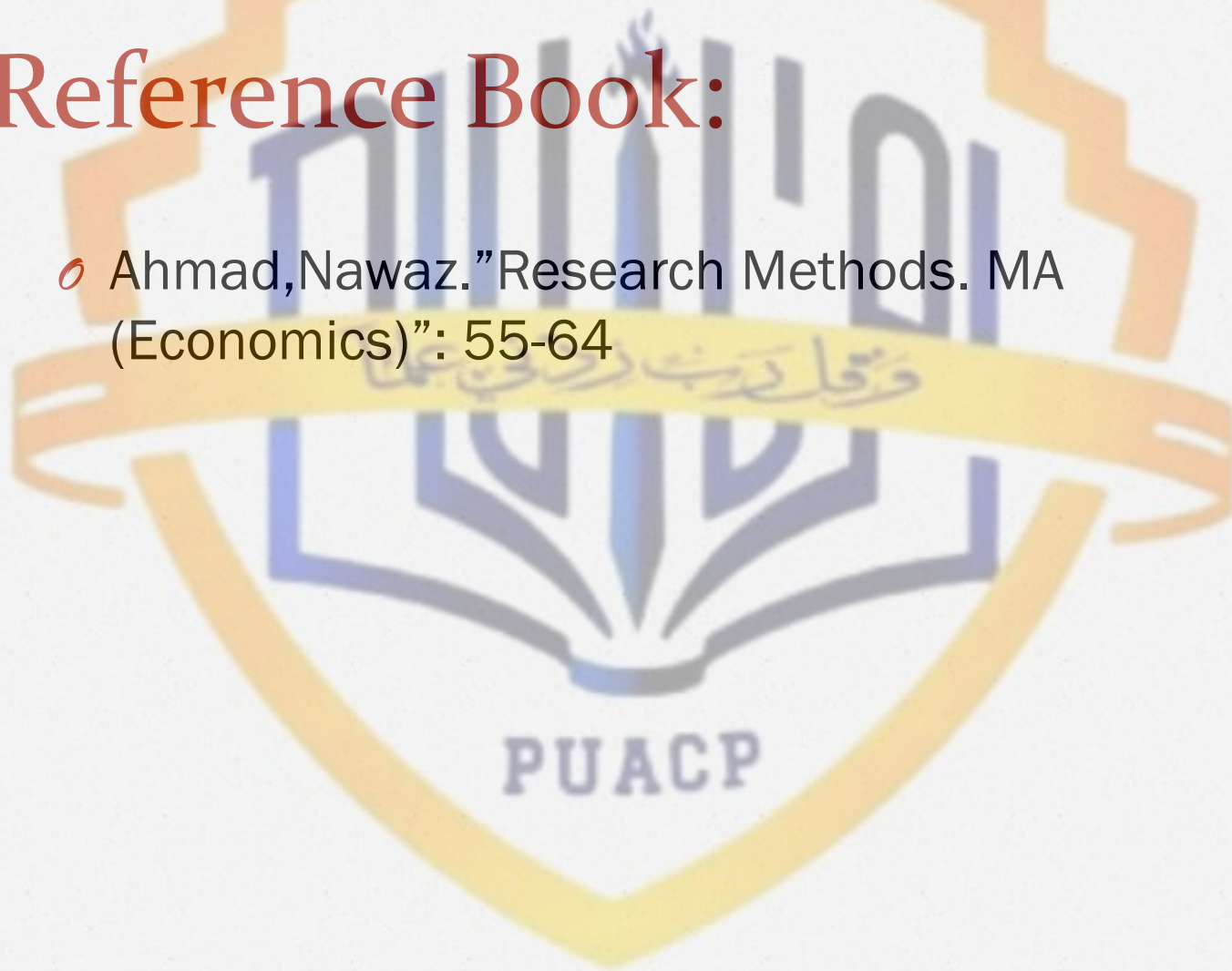
Components of a research design

Any investigator, before getting on to the job of planning a research design for a proposed study, should be clear about the following aspects:

- What to observe
- Whom to observe
- How to observe
- Why to observe
- How to record the observation
- How to analyze the observation
- What inferences can be drawn

Reference Book:

- Ahmad, Nawaz. "Research Methods. MA (Economics)": 55-64





Any Question ?

THANKS

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